> Table creation:

```
CREATE TABLE exoplanet_observations (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    Planet TEXT,
    Observation_Date TEXT,
    Normalized_Brightness REAL,
    Transit_Detected TEXT,
    Transit_Depth REAL,
    Orbital_Period_Days REAL,
    Transit_Duration_Hours REAL,
    Observation_Quality TEXT
);
```

> Insertion command and viewing table:

```
sqlite> INSERT INTO exoplanet_observations
   ...> (Planet, Observation_Date, Normalized_Brightness, Transit_Detected,
(x1...>  Transit_Depth, Orbital_Period_Days, Transit_Duration_Hours, Observation_Quality)
   ...> VALUES
   ...> ('Kepler-10 b', '2025-01-01', 0.998599, 'Yes', 0.0015, 0.8375, 2.2, 'Excellent'),
   ...> ('Kepler-10 b', '2025-01-01', 0.999972, 'No', 0.0000, 0.8375, 2.2, 'Fair'),
   ...> ('Kepler-10 b', '2025-01-01', 0.999778, 'No', 0.0000, 0.8375, 2.2, 'Excellent'),
   ...> ('Kepler-10 b', '2025-01-01', 0.998564, 'Yes', 0.0015, 0.8375, 2.2, 'Excellent'),
   ...> ('Kepler-10 b', '2025-01-02', 1.000316, 'No', 0.0000, 0.8375, 2.2, 'Good'),
   ...> ('Kepler-10 b', '2025-01-02', 1.000153, 'No', 0.0000, 0.8375, 2.2, 'Good'),
   ...> ('Kepler-10 b', '2025-01-02', 0.998384, 'Yes', 0.0015, 0.8375, 2.2, 'Good'),
   ...> ('Kepler-10 b', '2025-01-02', 0.999895, 'No', 0.0000, 0.8375, 2.2, 'Good'),
   ...> ('Kepler-10 b', '2025-01-03', 1.000048, 'No', 0.0000, 0.8375, 2.2, 'Fair'),
[  ...> ('Kepler-10 b', '2025-01-03', 0.998117, 'Yes', 0.0015, 0.8375, 2.2, 'Good');
[sqlite> SELECT * FROM exoplanet_observations;
1|Kepler-10 b|2025-01-01|0.998599|Yes|0.0015|0.8375|2.2|Excellent
2|Kepler-10 b|2025-01-01|0.999972|No|0.0|0.8375|2.2|Fair
3|Kepler-10 b|2025-01-01|0.999778|No|0.0|0.8375|2.2|Excellent
4|Kepler-10 b|2025-01-01|0.998564|Yes|0.0015|0.8375|2.2|Excellent
5|Kepler-10 b|2025-01-02|1.000316|No|0.0|0.8375|2.2|Good
6|Kepler-10 b|2025-01-02|1.000153|No|0.0|0.8375|2.2|Good
7|Kepler-10 b|2025-01-02|0.998384|Yes|0.0015|0.8375|2.2|Good
8|Kepler-10 b|2025-01-02|0.999895|No|0.0|0.8375|2.2|Good
9|Kepler-10 b|2025-01-03|1.000048|No|0.0|0.8375|2.2|Fair
10|Kepler-10 b|2025-01-03|0.998117|Yes|0.0015|0.8375|2.2|Good
```

> Find observations with transit depth greater than zero

```
sqlite> SELECT Planet, Transit_Depth, Orbital_Period_Days
   ...> FROM exoplanet_observations
   ...> WHERE Transit_Depth > 0
[  ...> ORDER BY Transit_Depth DESC;
Kepler-10 b|0.0015|0.8375
Kepler-10 b|0.0015|0.8375
Kepler-10 b|0.0015|0.8375
Kepler-10 b|0.0015|0.8375
```

> Find observations brighter than the average brightness

```
sqlite> SELECT Planet, Transit_Depth
   ...> FROM exoplanet_observations
   ...> WHERE Transit_Depth = (
(x1...>     SELECT MAX(Transit_Depth)
(x1...>     FROM exoplanet_observations
(x1...> );
Kepler-10 b|0.0015
Kepler-10 b|0.0015
Kepler-10 b|0.0015
Kepler-10 b|0.0015
sqlite>
```

> Find planets with at least one detected transit

```
sqlite> SELECT DISTINCT Planet
   ...> FROM exoplanet_observations
   ...> WHERE Planet IN (
(x1...>     SELECT Planet
(x1...>     FROM exoplanet_observations
(x1...>     WHERE Transit_Detected = 'Yes'
(x1...> );
Kepler-10 b
sqlite>
```

> Find observations whose brightness is above the average of their observation date

```
sqlite> SELECT o.Planet, o.Observation_Date, o.Normalized_Brightness
   ...> FROM exoplanet_observations o
   ...> WHERE o.Normalized_Brightness > (
(x1...>     SELECT AVG(i.Normalized_Brightness)
(x1...>     FROM exoplanet_observations i
(x1...>     WHERE i.Observation_Date = o.Observation_Date
(x1...> );
Kepler-10 b|2025-01-01|0.999972
Kepler-10 b|2025-01-01|0.999778
Kepler-10 b|2025-01-02|1.000316
Kepler-10 b|2025-01-02|1.000153
Kepler-10 b|2025-01-02|0.999895
Kepler-10 b|2025-01-03|1.000048
sqlite>
```

> **observations with a transit depth greater than the average transit depth**

```
sqlite> SELECT Planet, Transit_Depth
   ...> FROM exoplanet_observations
   ...> WHERE Transit_Depth > (
(x1...>     SELECT AVG(Transit_Depth)
(x1...>     FROM exoplanet_observations
(x1...> );
Kepler-10 b|0.0015
Kepler-10 b|0.0015
Kepler-10 b|0.0015
Kepler-10 b|0.0015
```

> Find observations where a transit was detected using `EXISTS`

```
sqlite> SELECT o.Planet, o.Observation_Date, o.Transit_Detected
   ...> FROM exoplanet_observations o
   ...> WHERE EXISTS (
(x1...>     SELECT 1
(x1...>     FROM exoplanet_observations i
(x1...>     WHERE i.id = o.id
(x1...>       AND i.Transit_Detected = 'Yes'
(x1...> );
Kepler-10 b|2025-01-01|Yes
Kepler-10 b|2025-01-01|Yes
Kepler-10 b|2025-01-02|Yes
Kepler-10 b|2025-01-03|Yes
sqlite>
```

ECommerce dataset

> creation of tables and insertion of data and viewing table

```
sqlite> CREATE TABLE ecommerce_cleaned (
(x1...>     Order_ID TEXT PRIMARY KEY,
(x1...>     Order_Date TEXT,
(x1...>     Customer_Name TEXT,
(x1...>     Product TEXT,
(x1...>     Category TEXT,
(x1...>     Quantity INTEGER,
(x1...>     Unit_Price REAL,
(x1...>     Discount_Pct REAL,
(x1...>     Payment_Method TEXT,
(x1...>     City TEXT,
(x1...>     Order_Status TEXT,
(x1...>     Shipping_Days INTEGER,
(x1...>     Gross_Amount REAL,
(x1...>     Discount_Amount REAL,
(x1...>     Net_Amount REAL,
(x1...>     Order_Value_Band TEXT
(x1...> );
sqlite> INSERT INTO ecommerce_cleaned VALUES
   ...> ('ORD-1001', '2025-01-01', 'Neha Sharma', 'Wireless Mouse', 'Electronics', 1, 799.000, 0, 'Upi', 'Hyderabad', 'Delivered', 1, 799.000, 0.000, 799.000, 'Medium'),
   ...>
   ...> ('ORD-1002', '2025-01-04', 'Neha Sharma', 'Bluetooth Speaker', 'Clothing', 2, 569.050, 5, 'Credit Card', 'Chennai', 'Processing', 2, 1138.100, 56.905, 1081.195, 'Medium'),
   ...>
   ...> ('ORD-1003', '2025-01-07', 'Ravi Kumar', 'Rice 5 Kg', 'Electronics', 1, 418.950, 10, 'Debit Card', 'Bengaluru', 'Returned', 3, 418.950, 41.895, 377.055, 'Low'),
   ...>
   ...> ('ORD-1004', '2025-01-10', 'Isha Reddy', 'Ballpoint Pens', 'Stationery', 3, 99.000, 15, 'Credit Card', 'Mumbai', 'Shipped', 4, 297.000, 44.550, 252.450, 'Low'),
   ...>
   ...> ('ORD-1005', '2025-01-13', 'Karan Patel', 'Desk Lamp', 'Home & Kitchen', 2, 1079.100, 20, 'Net Banking', 'Hyderabad', 'Cancelled', 5, 2158.200, 431.640, 1726.560, 'High'),
   ...>
   ...> ('ORD-1006', '2025-01-16', 'Ananya Rao', 'Bluetooth Speaker', 'Electronics', 4, 1499.000, 8, 'Upi', 'Delhi', 'Shipped', 6, 5996.000, 479.680, 5516.320, 'High'),
   ...>
   ...> ('ORD-1007', '2025-01-19', 'Vikram Singh', 'Coffee Mug', 'Home & Kitchen', 2, 236.550, 12, 'Credit Card', 'Kolkata', 'Processing', 7, 473.100, 56.772, 416.328, 'Low'),
   ...>
   ...> ('ORD-1008', '2025-01-22', 'Priya Nair', 'Usb-C Cable', 'Electronics', 5, 584.025, 0, 'Debit Card', 'Jaipur', 'Returned', 1, 2920.125, 0.000, 2920.125, 'High'),
   ...>
   ...> ('ORD-1009', '2025-01-25', 'Arjun Das', 'Running Shoes', 'Footwear', 2, 2199.000, 10, 'Cod', 'Hyderabad', 'Shipped', 2, 4398.000, 439.800, 3958.200, 'High'),
   ...>
   ...> ('ORD-1012', '2025-01-06', 'Unknown Customer', 'Rice 5 Kg', 'Groceries', 2, 379.050, 5, 'Credit Card', 'Mumbai', 'Processing', 5, 758.100, 37.905, 720.195, 'Medium');
sqlite> SELECT * FROM ecommerce_cleaned;
ORD-1001|2025-01-01|Neha Sharma|Wireless Mouse|Electronics|1|799.0|0.0|Upi|Hyderabad|Delivered|1|799.0|0.0|799.0|Medium
ORD-1002|2025-01-04|Neha Sharma|Bluetooth Speaker|Clothing|2|569.05|5.0|Credit Card|Chennai|Processing|2|1138.1|56.905|1081.195|Medium
ORD-1003|2025-01-07|Ravi Kumar|Rice 5 Kg|Electronics|1|418.95|10.0|Debit Card|Bengaluru|Returned|3|418.95|41.895|377.055|Low
ORD-1004|2025-01-10|Isha Reddy|Ballpoint Pens|Stationery|3|99.0|15.0|Credit Card|Mumbai|Shipped|4|297.0|44.55|252.45|Low
ORD-1005|2025-01-13|Karan Patel|Desk Lamp|Home & Kitchen|2|1079.1|20.0|Net Banking|Hyderabad|Cancelled|5|2158.2|431.64|1726.56|High
ORD-1006|2025-01-16|Ananya Rao|Bluetooth Speaker|Electronics|4|1499.0|8.0|Upi|Delhi|Shipped|6|5996.0|479.68|5516.32|High
ORD-1007|2025-01-19|Vikram Singh|Coffee Mug|Home & Kitchen|2|236.55|12.0|Credit Card|Kolkata|Processing|7|473.1|56.772|416.328|Low
ORD-1008|2025-01-22|Priya Nair|Usb-C Cable|Electronics|5|584.025|0.0|Debit Card|Jaipur|Returned|1|2920.125|0.0|2920.125|High
ORD-1009|2025-01-25|Arjun Das|Running Shoes|Footwear|2|2199.0|10.0|Cod|Hyderabad|Shipped|2|4398.0|439.8|3958.2|High
ORD-1012|2025-01-06|Unknown Customer|Rice 5 Kg|Groceries|2|379.05|5.0|Credit Card|Mumbai|Processing|5|758.1|37.905|720.195|Medium
```

> Calculate total revenue by city

```
sqlite> SELECT
   ...>     City,
   ...>     COUNT(Order_ID) AS Total_Orders,
   ...>     SUM(Quantity) AS Total_Units_Sold,
   ...>     ROUND(SUM(Net_Amount), 2) AS Total_Revenue
   ...> FROM ecommerce_cleaned
   ...> GROUP BY City
   ...> ORDER BY Total_Revenue DESC;
Hyderabad|3|5|6483.76
Delhi|1|4|5516.32
Jaipur|1|5|2920.13
Chennai|1|2|1081.19
Mumbai|2|5|972.64
Kolkata|1|2|416.33
Bengaluru|1|1|377.06
```

> Find product categories with above-average order value

```
sqlite> SELECT
   ...>     Order_ID,
   ...>     Customer_Name,
   ...>     Product,
   ...>     Net_Amount
   ...> FROM ecommerce_cleaned
   ...> WHERE Net_Amount > (
(x1...>     SELECT AVG(Net_Amount)
(x1...>     FROM ecommerce_cleaned
(x1...> )
   ...> ORDER BY Net_Amount DESC;
ORD-1006|Ananya Rao|Bluetooth Speaker|5516.32
ORD-1009|Arjun Das|Running Shoes|3958.2
ORD-1008|Priya Nair|Usb-C Cable|2920.125
```

>Find orders whose net amount is greater than the overall average

```
sqlite> SELECT
   ...>     Category,
   ...>     COUNT(*) AS Total_Orders,
   ...>     ROUND(AVG(Net_Amount), 2) AS Average_Order_Value
   ...> FROM ecommerce_cleaned
   ...> GROUP BY Category
   ...> HAVING AVG(Net_Amount) > (
(x1...>     SELECT AVG(Net_Amount)
(x1...>     FROM ecommerce_cleaned
(x1...> )
   ...> ORDER BY Average_Order_Value DESC;
Footwear|1|3958.2
Electronics|4|2403.13
```

>Find customers who spend more than the average customer

```
sqlite> SELECT
   ...>     Order_ID,
   ...>     Customer_Name,
   ...>     Product,
   ...>     Net_Amount
   ...> FROM ecommerce_cleaned
   ...> WHERE Net_Amount > (
(x1...>     SELECT AVG(Net_Amount)
(x1...>     FROM ecommerce_cleaned
(x1...> )
   ...> ORDER BY Net_Amount DESC;
ORD-1006|Ananya Rao|Bluetooth Speaker|5516.32
ORD-1009|Arjun Das|Running Shoes|3958.2
ORD-1008|Priya Nair|Usb-C Cable|2920.125
```

> Categorize orders by shipping performance

```
sqlite> SELECT
   ...>     Customer_Name,
   ...>     ROUND(SUM(Net_Amount), 2) AS Total_Spending
   ...> FROM ecommerce_cleaned
   ...> GROUP BY Customer_Name
   ...> HAVING SUM(Net_Amount) > (
(x1...>     SELECT AVG(Customer_Total)
(x1...>     FROM (
(x2...>         SELECT SUM(Net_Amount) AS Customer_Total
(x2...>         FROM ecommerce_cleaned
(x2...>         GROUP BY Customer_Name
(x2...>     )
(x1...> )
   ...> ORDER BY Total_Spending DESC;
Ananya Rao|5516.32
Arjun Das|3958.2
Priya Nair|2920.13
```

> **Rank products by revenue within each category**

```
sqlite> SELECT
   ...>     Order_ID,
   ...>     Customer_Name,
   ...>     Order_Status,
   ...>     Shipping_Days,
   ...>     CASE
   ...>         WHEN Order_Status = 'Returned' THEN 'Returned Order'
   ...>         WHEN Order_Status = 'Cancelled' THEN 'Cancelled Order'
   ...>         WHEN Shipping_Days <= 2 THEN 'Fast Shipping'
   ...>         WHEN Shipping_Days <= 5 THEN 'Normal Shipping'
   ...>         ELSE 'Slow Shipping'
   ...>     END AS Shipping_Category
   ...> FROM ecommerce_cleaned
   ...> ORDER BY Shipping_Days DESC;
ORD-1007|Vikram Singh|Processing|7|Slow Shipping
ORD-1006|Ananya Rao|Shipped|6|Slow Shipping
ORD-1005|Karan Patel|Cancelled|5|Cancelled Order
ORD-1012|Unknown Customer|Processing|5|Normal Shipping
ORD-1004|Isha Reddy|Shipped|4|Normal Shipping
ORD-1003|Ravi Kumar|Returned|3|Returned Order
ORD-1002|Neha Sharma|Processing|2|Fast Shipping
ORD-1009|Arjun Das|Shipped|2|Fast Shipping
ORD-1001|Neha Sharma|Delivered|1|Fast Shipping
ORD-1008|Priya Nair|Returned|1|Returned Order
```

> Rank products by revenue within each category

```
sqlite> SELECT
   ...>     Category,
   ...>     Product,
   ...>     Total_Revenue,
   ...>     RANK() OVER (
(x1...>         PARTITION BY Category
(x1...>         ORDER BY Total_Revenue DESC
(x1...>     ) AS Product_Rank
   ...> FROM (
(x1...>     SELECT
(x1...>         Category,
(x1...>         Product,
(x1...>         ROUND(SUM(Net_Amount), 2) AS Total_Revenue
(x1...>     FROM ecommerce_cleaned
(x1...>     GROUP BY Category, Product
(x1...> ) AS Product_Sales
   ...> ORDER BY Category, Product_Rank;
Clothing|Bluetooth Speaker|1081.19|1
Electronics|Bluetooth Speaker|5516.32|1
Electronics|Usb-C Cable|2920.13|2
Electronics|Wireless Mouse|799.0|3
Electronics|Rice 5 Kg|377.06|4
Footwear|Running Shoes|3958.2|1
Groceries|Rice 5 Kg|720.2|1
Home & Kitchen|Desk Lamp|1726.56|1
Home & Kitchen|Coffee Mug|416.33|2
Stationery|Ballpoint Pens|252.45|1
```

> Calculate monthly revenue and running revenue

```
sqlite> WITH Monthly_Sales AS (
(x1...>     SELECT
(x1...>         strftime('%Y-%m', Order_Date) AS Sales_Month,
(x1...>         ROUND(SUM(Net_Amount), 2) AS Monthly_Revenue
(x1...>     FROM ecommerce_cleaned
(x1...>     GROUP BY strftime('%Y-%m', Order_Date)
(x1...> )
   ...> SELECT
   ...>     Sales_Month,
   ...>     Monthly_Revenue,
   ...>     ROUND(
(x1...>         SUM(Monthly_Revenue) OVER (
(x2...>             ORDER BY Sales_Month
(x2...>             ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
(x2...>         ),
(x1...>         2
(x1...>     ) AS Running_Revenue
   ...> FROM Monthly_Sales
   ...> ORDER BY Sales_Month;
2025-01|17767.43|17767.43
```